SAKSHAM BANSAL

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SUMMARY

Software Engineer with internship research experience and 6+ shipped projects in Python, C++, Java. Goldman Sachs India Hackathon Semi Finale (1st of 5,000+); AIR 5 in IMC Prosperity Trading Challenge in Round 3(22,000+ teams). Skilled in backend systems, REST APIs and machine learning.

EDUCATION

Indraprastha Institute of Information Technology Delhi (IIITD) CGPA-7.82 Delhi, India
B.Tech in Computer Science and Applied Mathematics Aug 2023 – May 2027
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, OOP, Probability & Statistics, Machine Learning, Reinforcement Learning

SKILLS

Languages: C, C++, Python, Java, SQL (PostgreSQL, MySQL)
Tools & Frameworks: Git, GitHub, Linux, Redis, Pandas, NumPy, Streamlit, pytest, JUnit, LibGDX
Core Competencies: Software Development, Algorithms, Data Structures, REST APIs, Unit Testing, System Design, Machine Learning, Reinforcement Learning, NLP, OOP

WORK EXPERIENCE

Sengupta Lab, IIIT Delhi – *Software Developer (Research)* May 2025 – Aug 2025
- Owned end-to-end delivery of an Agentic NL2SQL framework enabling natural-language queries over **1M+ biomedical records**.
- Accelerated retrieval with parallel multi-database pipelines, achieving **3x faster execution** across **10K+ queries**.
- Automated preprocessing across 6 data sources with a cross-functional team, cutting manual effort by **70%**.

OPEN SOURCE EXPERIENCE

Open Source Contributor – CHAOSS Augur (Linux Foundation) – *Python, GitHub API* June 2025 – Jul 2025
- Contributed production-ready Python code to CHAOSS Augur, implementing HTTP 404 handling and repository move detection to improve collector reliability.
- Collaborated with Linux Foundation maintainers through multiple review cycles, improving exception handling, logging, and production collector behavior.

PROJECTS

Multimedia Compression and Emotion Recognition – *Python, NumPy, Pandas, Librosa, Pillow* Aug 2025 – Sep 2025
- Benchmarked audio and image compression using **PSNR** and **SSIM** across large multimedia datasets.
- Developed speech- and facial-image-based emotion recognition pipelines, evaluating performance using accuracy metrics and confusion matrices.
- Produced comparative visualizations to analyze compression quality and classification performance.

DocBot – AI Documentation Chatbot – *Python, Streamlit, Gemini API, LanceDB* May 2025 – Jun 2025
- Designed a semantic-search chatbot achieving **sub-1s latency** and **95% retrieval accuracy**.
- Launched a secure Streamlit Cloud deployment with real-time indexing and a responsive interface.

Agricultural Marketplace Platform – *TypeScript, PostgreSQL, WebSockets* Jan 2025 – Apr 2025
- Led a **4-person team** to ship a full-stack marketplace with **50+ workflows** and role-based authentication.
- Optimized PostgreSQL schemas for **10K+ records**, cutting search latency by **40%** with real-time WebSocket feeds.

Angry Birds Game Clone – *Java, LibGDX, Box2D* Sep 2024 – Dec 2024
- Architected a physics-based game engine using OOP and the Strategy Pattern for extensible gameplay logic.
- Raised unit-test coverage to **95%** with JUnit and produced **5+ UML diagrams** for maintainable architecture.

Linux Shell – *C, Operating Systems* Sep 2024 – Dec 2024
- Programmed a UNIX-like shell with history, piping, signal handling, and **10+ concurrent processes**.

ACHIEVEMENTS

- **IMC Prosperity 4 Trading Challenge 2026** – Achieved AIR 5 and Global Rank 65 of **22,000+ teams** designing algorithmic trading strategies.
- **Goldman Sachs India Hackathon 2025** – Ranked 1st of **5,000+ participants**; advanced to the national Semi-Finale.
- **Y Combinator Startup School India 2026** – Selected from **25,000+ applicants** for the Bangalore cohort.
- **Codeforces Specialist** – Earned max rating **1444**; solved **450+ problems** across Codeforces, LeetCode, and CodeChef.
- **JEE Mains 2023** – Scored in the top **1% of 1.5M+ candidates** nationwide.